NAME: _____ Geometry (FL B.E.S.T.)

1.5 Conditional Statements

Every theorem in geometry is really a conditional statement in disguise, so learning to dissect "if p, then q" into its converse, inverse, and contrapositive is what makes later proofs readable. Beyond just rewriting a sentence, this section asks you to judge *truth* — proving a statement false with a single counterexample, or recognizing when a conditional and its contrapositive must share a truth value while the converse might not. It also introduces the two rules that power two-column proofs: the Law of Detachment and the Law of Syllogism, along with the classic converse-error trap of "affirming the consequent."

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. "If n is even, then n is divisible by 2." Hypothesis?

2. Same statement. Conclusion?

3. If a polygon is a triangle, then it has 3 sides. Converse?

4. If an angle is right, then it measures $90°$. Inverse?

5. If two lines are parallel, then they do not intersect. Contrapositive?

6. Which is logically equivalent to a conditional?

7. If $x^2 = 9$, then $x = 3$. Counterexample?

8. When is a biconditional true?

9. True conditional: If a figure is a square, then it is a rhombus. Is the converse always true?

10. Hypothesis: angle is straight. Conclusion: measure is $180°$. Write if–then.

11. If $AB = CD$, then $\overline{AB} \cong \overline{CD}$. Inverse?

12. If $m\angle A = 90°$, then $\angle A$ is right. Contrapositive?

13. Symbolic form of "if p then q"?

14. (1) If a polygon is a regular hexagon, then all its sides are congruent. (2) If all of a polygon's sides are congruent, then it is equilateral. Combine using the Law of Syllogism.

15. If two angles form a linear pair, then they are supplementary. $\angle 1$ and $\angle 2$ are a linear pair, and $m\angle 1 = 112°$. Find $m\angle 2$ and name the law used.

✓ Answer Key

1. n is even
2. n is divisible by 2
3. If a polygon has 3 sides, then it is a triangle
4. If an angle is not right, then it does not measure $90°$
5. If two lines intersect, then they are not parallel
6. Contrapositive
7. $x = -3$
8. When both the conditional and its converse are true
9. No
10. If an angle is straight, then its measure is $180°$
11. If $AB \not\equiv CD$, then $\overline{AB} \ncong \overline{CD}$
12. If $\angle A$ is not right, then $m\angle A \not\equiv 90°$
13. $p \rightarrow q$
14. If a polygon is a regular hexagon, then it is equilateral
15. $m\angle 2 = 68°$; Law of Detachment